

Errata for Paper I

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June 2026

This is the errata for the printed version of Aguiar, M., “Learning differentiable simulation models of control systems” (2026), PhD Thesis, KTH Royal Institute of Technology.

1. Corrections to Table 1 (i), see below. Specifically, the ‘steps’ in BFGS for $\hat{\theta}_0 = 0.0$ and $\hat{\theta}_0 = 1.5$, respectively, and the ‘error’ for $\hat{\theta} = 0.0$ for method T5. Old entries are shown in red after the updated number.
2. In Section 6, “the encoder and decoder have 4 and 3 layers, respectively, with 128 nodes each”, should read “the encoder and decoder have 3 layers with 128 nodes”.

(i) $M = 60$

$\hat{\theta}_0$	Gradient descent				BFGS			
	Error		Steps		Error		Steps	
	F1	T5	F1	T5	F1	T5	F1	T5
0.0	4.21e-3	8.85	9	2	4.14e-3	4.95e-6 / 4.14e-3	14 / 12	26 / 12
0.5	4.06e-3	9.75e-5	13	12	4.15e-3	7.75e-7	9	13
1.5	4.29e-3	4.36e-5	12	13	4.13e-3	2.98e-6	11 / 9	11 / 11
2.0	4.26e-3	4.80e-5	26	26	4.14e-3	7.75e-6	9	10
2.5	1.20	1.24	32	24	4.13e-3	7.33e-6	8	9
3.0	1.20	1.87	90	23	4.14e-3	7.87e-6	12	9
3.5	2.50	2.07	1	33	2.14	7.93e-6	9	10

Table 1: Estimation of the Van der Pol damping parameter for different initial points $\hat{\theta}_0$ using gradient descent and BFGS.